

CBSE Class 12 Chemistry Previous Year Question Paper – Set 56/1/1

Time Allowed: 3 hours

Maximum Marks: 70

General Instructions

- All questions are compulsory.
- Internal choice is provided wherever applicable.
- Questions follow the current CBSE syllabus.

SECTION A – Objective Questions

1. Which of the following has the highest melting point?

- (a) NaCl
- (b) KCl
- (c) MgCl_2
- (d) CaCl_2

2. The bond order in O_2^+ ion is

- (a) 2.5
- (b) 3.0
- (c) 1.5
- (d) 2.0

3. The geometry of $[\text{Ni}(\text{CN})_4]^{2-}$ ion is

- (a) tetrahedral
- (b) square planar
- (c) octahedral
- (d) linear

4. Which among the following compounds exhibits enantiomerism?

- (a) 1-Butanol
- (b) 2-Butanol
- (c) 2-Methyl-1-propanol
- (d) Tert-butanol

5. The oxidation number of Fe in $K_4[Fe(CN)_6]$ is

- (a) 2
- (b) 4
- (c) 0
- (d) 6

6. Bakelite is obtained by the reaction between

- (a) formaldehyde and phenol
- (b) urea and formaldehyde
- (c) butadiene and styrene
- (d) ethylene and benzene

7. The rate constant of a first order reaction is 0.693 min^{-1} . Its half-life is

- (a) 1.0 min
- (b) 0.5 min
- (c) 2.0 min
- (d) 1.5 min

8. The main ore of aluminium is

- (a) Magnetite
- (b) Cuprite
- (c) Cassiterite
- (d) Bauxite

9. Which ion shows maximum hydration energy?

- (a) Na^+
- (b) Mg^{2+}
- (c) K^+
- (d) Ca^{2+}

10. Which one of the following is a biodegradable polymer?

- (a) Polystyrene
- (b) PHBV
- (c) PVC
- (d) Bakelite

11. The correct order of acidic strength of nitrophenols is

- (a) ortho > para > meta
- (b) para > ortho > meta
- (c) meta > para > ortho
- (d) ortho > meta > para

12. Ammonia reacts with Cu^{2+} to give a deep blue coloured complex. This reaction is an example of:

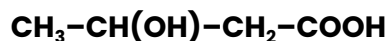
- (a) Redox reaction
- (b) Precipitation reaction
- (c) Complex formation
- (d) Double displacement

13. Assertion (A): Boiling point of water is higher than that of ammonia.

Reason (R): Water forms stronger hydrogen bonds than ammonia.

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true, R is false.
- (d) A is false, R is true.

14. What is the IUPAC name of the following compound?



- (a) 3-Hydroxybutanoic acid
- (b) 2-Hydroxybutanoic acid
- (c) Butan-3-ol-1-oic acid
- (d) 2-Hydroxypropanoic acid

15. Which of the following acts as an anticoagulant in blood?

- (a) Vitamin K
- (b) Heparin
- (c) Insulin
- (d) Thyroxin

16. The increasing order of basicity of the following is:

- (a) Aniline < Ammonia < Ethylamine
- (b) Ammonia < Aniline < Ethylamine
- (c) Ethylamine < Aniline < Ammonia
- (d) Aniline < Ethylamine < Ammonia

17. Which one among the following is a non-reducing sugar?

- (a) Glucose
- (b) Maltose
- (c) Sucrose
- (d) Lactose

18. The van't Hoff factor for a solution of K_2SO_4 in water is expected to be

- (a) 1
- (b) 2
- (c) 3
- (d) 4

19. The reaction of ethyl bromide with aqueous KOH produces

- (a) Ethanol
- (b) Ethene
- (c) Acetylene
- (d) Diethyl ether

SECTION B – Short Answer Questions (2 marks)

20. Define F-centres. How does their presence affect the color of alkali halides?

21. Why does increasing the temperature generally increase the rate of a chemical reaction?

22. State Raoult's law for ideal solutions. Give one example of an ideal solution.

23. Write the cell reaction and calculate the EMF of the following cell at 25°C:



24. Explain why transition metals form colored compounds.

25. Give the equation for the preparation of acetaldehyde from ethanol. Name the catalyst used.

26. Differentiate between thermoplastic and thermosetting polymers with suitable examples.

27. How does the presence of non-volatile solute affect the boiling point of a solvent? Name this phenomenon.

28. Although NH_3 has higher molar mass than PH_3 , it has lower boiling point. Explain.

29. Write the relation between Gibbs energy and cell potential for an electrochemical cell.
30. The rate equation for a reaction is: $\text{rate} = k [A]^1[B]^2$. What is the order of the reaction?
31. Write one example each of synthetic and natural polymers.
32. Give the IUPAC name for $(\text{CH}_3)_3\text{CNH}_2$.
33. Explain the difference between competitive and non-competitive enzyme inhibition.
34. Why is potassium permanganate considered a self-indicator in redox titrations?
35. Write two differences between lyophilic and lyophobic sols.
36. State the function of DNA in the synthesis of proteins.
37. Draw the structure of 2,4-dinitrophenol.
38. What are semiconductors? Give one example.
39. Arrange the following in increasing order of bond length: N_2 , O_2 , F_2

SECTION C – 3-Mark Questions

40. Describe the steps involved in the extraction of copper from its sulphide ore.
41. Give the mechanism of nucleophilic substitution reaction of methyl bromide with aqueous sodium hydroxide.

42. How will you distinguish between primary, secondary, and tertiary alcohols using Lucas reagent?

43. Explain the lanthanide contraction and its consequences.

44. Explain the principle and process of froth flotation method used for concentration of ore.

45. Describe the importance of green chemistry in chemical synthesis.

46. What is the difference between order and molecularity of a reaction? Give an example for each.

47. Explain the term 'peptide bond' with the help of a diagram.

48. Mention the main postulates of the valence bond theory as applied to transition metal complexes.

49. Write the chemical equations involved in the reaction of glucose with: (i) bromine water, (ii) HI.

SECTION D – Case Study / Source-Based Questions

50. Read the passage and answer the questions:

A student dissolves sodium chloride in water, cools the solution, and observes some white powder precipitating out. She filters and collects this powder, also measuring the conductivity before and after precipitation.

(a) Why does precipitation of sodium chloride occur upon cooling?

(b) How will conductivity change after precipitation and why?

51. The following data were recorded for the reaction: $2\text{NO}_2 \rightarrow 2\text{NO} + \text{O}_2$

Initial concentration of NO_2 : 0.5 M; Initial rate: $1.0 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$

Initial concentration of NO_2 : 1.0 M; Initial rate: $4.0 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$

- (a) Determine the order of the reaction with respect to NO_2 .
- (b) Explain why the rate changes as observed.

52. After adding excess NaOH to an unknown metal salt solution, a reddish-brown precipitate forms. Upon adding NH_4Cl , the precipitate dissolves.

- (a) Which metal ion is likely present and why?
- (b) What is the role of NH_4Cl in dissolving the precipitate?

SECTION E – Long Answer Questions (5 marks)

53. Discuss the principle, working, and applications of the electrolysis of molten sodium chloride. Also explain the reactions at both electrodes and the industrial significance of this process.

54. Explain the structural features, preparation, and uses of DDT. Discuss its impact on the environment, and state the reasons for restricting its use despite its efficacy as an insecticide.

55. Describe the mechanism of action of enzymes as biological catalysts. Include an explanation of active site, enzyme-substrate complex, specificity, and examples from metabolism where enzymes play a critical role.

Answer Key – CBSE Class 12 Chemistry – Set 56/1/1

1. (a)
2. (a)
3. (b)
4. (b)
5. (a)
6. (a)
7. (a)
8. (d)
9. (b)
10. (b)
11. (a)
12. (c)
13. (a)
14. (a)
15. (b)
16. (a)
17. (c)
18. (c)
19. (a)

20. F-centres are vacancies in the crystal lattice occupied by electrons; their presence imparts color to alkali halides.
21. Increase in temperature raises kinetic energy of molecules, leading to more effective collisions and thus higher reaction rate.
22. Raoult's law states that the partial vapor pressure of each component in an ideal solution is proportional to its mole fraction; example: benzene and toluene solution.
23. $\text{Zn(s)} + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Cu(s)}$; EMF ≈ 1.10 V.
24. Transition metals form colored compounds due to d-d electron transitions when exposed to light.
25. $\text{CH}_3\text{CH}_2\text{OH} + [\text{O}] \rightarrow \text{CH}_3\text{CHO} + \text{H}_2\text{O}$; catalyst: acidified $\text{K}_2\text{Cr}_2\text{O}_7$ or PCC.
26. Thermoplastics soften on heating and can be reshaped; e.g. PVC. Thermosetting polymers harden permanently after molding; e.g. Bakelite.
27. Boiling point of solvent increases due to non-volatile solute; this is called elevation of boiling point.
28. NH_3 forms strong hydrogen bonds, but has lower boiling point than PH_3 due to greater molecular interactions in PH_3 .
29. $\Delta G = -nFE_{\text{cell}}$ relates Gibbs energy and cell potential.
30. Order of the reaction is $1 + 2 = 3$.
31. Synthetic polymer: nylon; Natural polymer: cellulose.
32. 2-amino-2-methylpropane.
33. Competitive inhibition: inhibitor competes with substrate for active site; non-competitive inhibition: inhibitor binds elsewhere, changing enzyme shape.
34. KMnO_4 acts as its own indicator by changing color as it reacts and is reduced in redox titrations.

35. Lyophilic sols are easily prepared and reversible; lyophobic sols are not, and irreversible.
36. DNA carries genetic information and serves as a template for protein synthesis via transcription and translation.
37. Structure shows benzene ring with OH at 1-position and NO₂ at 2- and 4-positions.
38. Semiconductors have conductivity between metals and insulators; e.g. silicon.
39. Order: N₂ < O₂ < F₂.
40. Copper extraction: Ore is concentrated, roasted to oxide, then reduced using carbon; blister copper obtained by refining.
41. Methyl bromide reacts with NaOH via S_N2 mechanism, forming methanol and NaBr; involves one-step nucleophilic attack.
42. Lucas test: Primary alcohols react slowly, secondary-moderately fast, tertiary-immediately with Lucas reagent, forming turbidity at different rates.
43. Lanthanide contraction is the gradual decrease in ionic radius among lanthanides; causes similarities between 4d and 5d transition series.
44. Froth flotation involves mixing crushed ore with oil and water; air bubbles lift ore particles to surface for collection.
45. Green chemistry aims to minimize environmental impact by designing safer chemicals and reducing waste in synthesis.
46. Order is the sum of powers of concentration in rate law; molecularity refers to the number of reactant molecules in elementary step.
47. A peptide bond is formed between the carboxyl group of one amino acid and the amino group of another, releasing water; -CO-NH-.

48. Valence bond theory postulates hybridization of orbitals, formation of coordinate bonds, and predicts geometry of complexes.

49. (i) $\text{Glucose} + \text{Br}_2 \rightarrow \text{Gluconic acid}$ (ii) $\text{Glucose} + \text{HI} \rightarrow \text{n-Hexane}$.

50. (a) Solubility of NaCl decreases on cooling, causing precipitation. (b) Conductivity decreases because fewer ions remain in solution after precipitation.

51. (a) Order is 2, rate quadruples when $[\text{NO}_2]$ doubles. (b) The rate increases as concentration increases due to dependence on second power of reactant.

52. (a) The ion is likely Fe^{3+} ; reddish-brown ppt indicates ferric hydroxide. (b) NH_4Cl dissolves the ppt by shifting equilibrium and preventing hydrolysis.

53. Electrolysis of molten NaCl involves Na^+ reduced to Na at cathode, Cl^- oxidized to Cl_2 at anode; used industrially for sodium and chlorine production; reactions: $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$; $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$.

54. DDT has a chlorinated hydrocarbon structure, prepared by condensing chloral and chlorobenzene; used as insecticide but accumulates in food chain, harming wildlife, leading to bans.

55. Enzymes act by binding substrates at their active sites, forming enzyme-substrate complexes; show specificity, accelerate biological reactions; examples: amylase, DNA polymerase; vital for metabolism.